

Assignment 7: GLMs (Linear Regressios, ANOVA, & t-tests)

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A07_GLMs.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

!! AI generated code should NOT be used in this assignment

Set up your session

1. Set up your session. Check your working directory. Load the tidyverse, agricolae and other needed packages. Import the *raw* NTL-LTER raw data file for chemistry/physics (NTL-LTER_Lake_ChemistryPhysics_Raw.csv). Set date columns to date objects.
2. Build a ggplot theme and set it as your default theme.

#1

```
library(tidyverse);library(here);library(cowplot);library(ggplot2);library(ggthemes);library(agricolae)

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr    1.5.1
## v ggplot2     4.0.0      v tibble     3.2.1
## v lubridate  1.9.4      v tidyr      1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
## here() starts at /home/guest/EDE_Fall2025
##
##
```

```
## Attaching package: 'cowplot'
##
##
## The following object is masked from 'package:lubridate':
##
##     stamp
##
##
## Attaching package: 'ggthemes'
##
##
## The following object is masked from 'package:cowplot':
##
##     theme_map
```

```
here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
Lake_Chemistry <- read.csv(
  file = here("Data/Raw/NTL-LTER_Lake_ChemistryPhysics_Raw.csv"),
  stringsAsFactors = TRUE
)
```

```
class(Lake_Chemistry$sampleddate)
```

```
## [1] "factor"
```

```
Lake_Chemistry$sampleddate <- mdy(Lake_Chemistry$sampleddate)
```

```
#2
```

```
mytheme <- theme_classic(base_size = 14) +
  theme(axis.text = element_text(color = "navy"),
        legend.position = "top")
theme_set(mytheme)
```

Simple regression

Our first research question is: Does mean lake temperature recorded during July change with depth across all lakes?

3. State the null and alternative hypotheses for this question: > Answer: H0: The difference between the sample mean at varying depths is 0. Ha: The difference is not 0, it is greater than or less than 0.
4. Wrangle your NTL-LTER dataset with a pipe function so that the records meet the following criteria:
 - Only dates in July.
 - Only the columns: `lakename`, `year4`, `daynum`, `depth`, `temperature_C`
 - Only complete cases (i.e., remove NAs)

5. Visualize the relationship among the two continuous variables with a scatter plot of temperature by depth. Add a smoothed line showing the linear model, and limit temperature values from 0 to 35 °C. Make this plot look pretty and easy to read.

```
#4
Lake_Chemistry_Sub <- Lake_Chemistry %>%
  filter(daynum >= 183 & daynum <= 213) %>%
  select(c(lakename, year4, daynum, depth, temperature_C)) %>%
  na.omit()

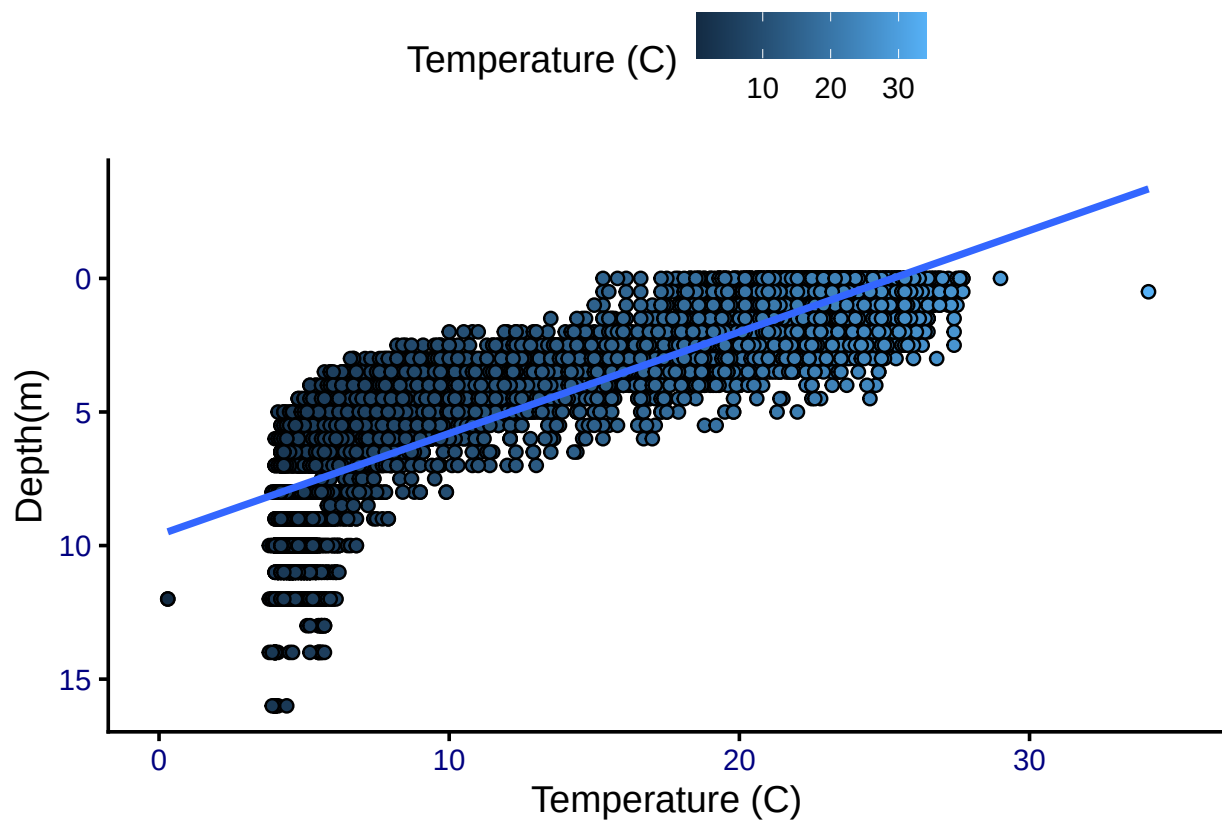
#5

temp_depth_plot <- ggplot(Lake_Chemistry_Sub, aes(x = temperature_C, y = depth, fill = temperature_C)) +
  geom_point(shape = 21) +
  geom_smooth(method = "lm") +
  scale_y_reverse() +
  xlim(0,35) +
  labs(x= "Temperature (C)", y="Depth(m)") +
  scale_fill_continuous(name = "Temperature (C)")

print(temp_depth_plot)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: The following aesthetics were dropped during statistical transformation: fill.
## i This can happen when ggplot fails to infer the correct grouping structure in
##   the data.
## i Did you forget to specify a 'group' aesthetic or to convert a numerical
##   variable into a factor?
```



6. Interpret the figure. What does it suggest with regards to the response of temperature to depth? Do the distribution of points suggest about anything about the linearity of this trend?

Answer: Based on the slope of the lm line, it shows a strong correlation where as depth increases, temperature decreases. The point distribution echo the thermocline likely present in the lake.

7. Perform a linear regression to test the relationship and display the results.

```
#7

temp_depth_regression <- lm(data = Lake_Chemistry_Sub, temperature_C ~ depth)

summary(temp_depth_regression)

##
## Call:
## lm(formula = temperature_C ~ depth, data = Lake_Chemistry_Sub)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.5606  -3.0380   0.0872   2.9872  13.4706
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept) 21.98318    0.06840    321.4    <2e-16 ***
## depth      -1.94086    0.01179   -164.7    <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.852 on 9671 degrees of freedom
## Multiple R-squared:  0.7371, Adjusted R-squared:  0.7371
## F-statistic: 2.712e+04 on 1 and 9671 DF,  p-value: < 2.2e-16
```

8. Interpret your model results in words. Include how much of the variability in temperature is explained by changes in depth, the degrees of freedom on which this finding is based, and the statistical significance of the result. Also mention how much temperature is predicted to change for every 1m change in depth.

Answer: We would expect temperature to drop by -1.94C for each 1m change in depth, as represented by the slope of the regression line. This is based on 9,671 degrees of freedom and the R-square value shows that 73.7% of variation in temperature is explained by the depth of the measurement. The p-value is very small, so we will reject the null hypothesis and say there is a statistically significant relationship between depth and temperature.

Multiple regression

Let's tackle a similar question from a different approach. Here, we want to explore what might the best set of predictors for lake temperature in July across the monitoring period at the North Temperate Lakes LTER.

9. Run an AIC to determine what set of explanatory variables (year4, daynum, depth) is best suited to predict temperature.
10. Run a multiple regression on the recommended set of variables.

#9

```
Temp_AIC <- lm(data = Lake_Chemistry_Sub, temperature_C ~ depth + year4 + daynum)
step(Temp_AIC)
```

```
## Start:  AIC=25998.22
## temperature_C ~ depth + year4 + daynum
##
##           Df Sum of Sq    RSS   AIC
## <none>                 142056 25998
## - year4    1         201 142257 26010
## - daynum   1         1237 143293 26080
## - depth    1        402549 544605 38995
##
## Call:
## lm(formula = temperature_C ~ depth + year4 + daynum, data = Lake_Chemistry_Sub)
##
## Coefficients:
## (Intercept)      depth      year4      daynum
##   -18.19700    -1.94133     0.01611     0.04024
```

#10

```
Temp_model <- lm(data = Lake_Chemistry_Sub, temperature_C ~ depth + year4 + daynum)
summary(Temp_model)
```

```
##
## Call:
## lm(formula = temperature_C ~ depth + year4 + daynum, data = Lake_Chemistry_Sub)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.6857 -3.0267  0.1055  2.9937 13.6038
##
## Coefficients:
##              Estimate Std. Error  t value Pr(>|t|)
## (Intercept) -18.196998   8.741236   -2.082 0.037392 *
## depth        -1.941328   0.011728 -165.528 < 2e-16 ***
## year4         0.016113   0.004353   3.701 0.000216 ***
## daynum        0.040237   0.004385   9.176 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.833 on 9669 degrees of freedom
## Multiple R-squared:  0.7398, Adjusted R-squared:  0.7397
## F-statistic: 9162 on 3 and 9669 DF,  p-value: < 2.2e-16
```

11. What is the final set of explanatory variables that the AIC method suggests we use to predict temperature in our multiple regression? How much of the observed variance does this model explain? Is this an improvement over the model using only depth as the explanatory variable?

Answer: The AIC doesn't recommend removing any variables from the regression. The new model explains 73.9% of variance now (an improvement from 73.7%). This is a small improvement from the singular variable regression.

Analysis of Variance

12. Now we want to see whether the different lakes have, on average, different temperatures in the month of July. Run an ANOVA test to complete this analysis. (No need to test assumptions of normality or similar variances.) Create two sets of models: one expressed as an ANOVA models and another expressed as a linear model (as done in our lessons).

#12

```
# Format ANOVA as aov
Lake_Chemistry_ANOVA <- aov(data = Lake_Chemistry_Sub, temperature_C ~ lakename)
summary(Lake_Chemistry_ANOVA)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## lakename      8 22188  2773.5   51.18 <2e-16 ***
## Residuals    9664 523706    54.2
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Format ANOVA as lm
Lake_Chemistry_ANOVA2 <- lm(data = Lake_Chemistry_Sub, temperature_C ~ lakename)
summary(Lake_Chemistry_ANOVA2)
```

```
##
## Call:
## lm(formula = temperature_C ~ lakename, data = Lake_Chemistry_Sub)
##
## Residuals:
```

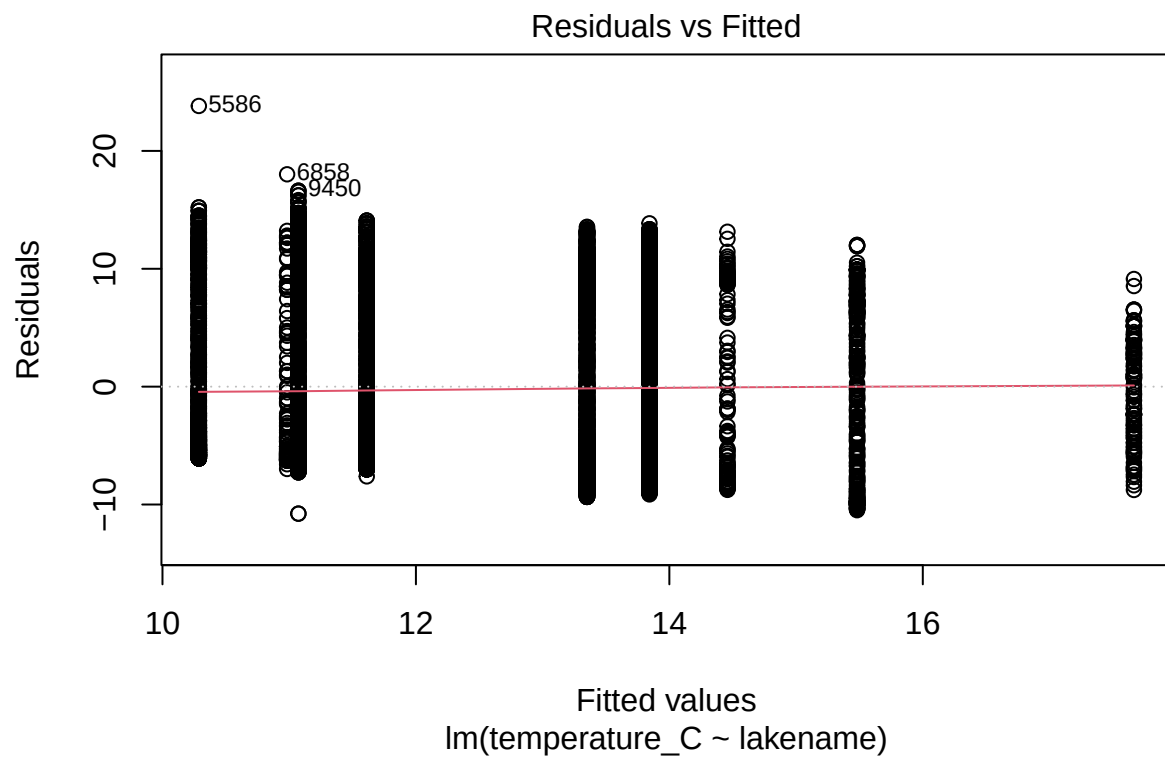
	Min	1Q	Median	3Q	Max
	-10.773	-6.612	-2.673	7.657	23.813

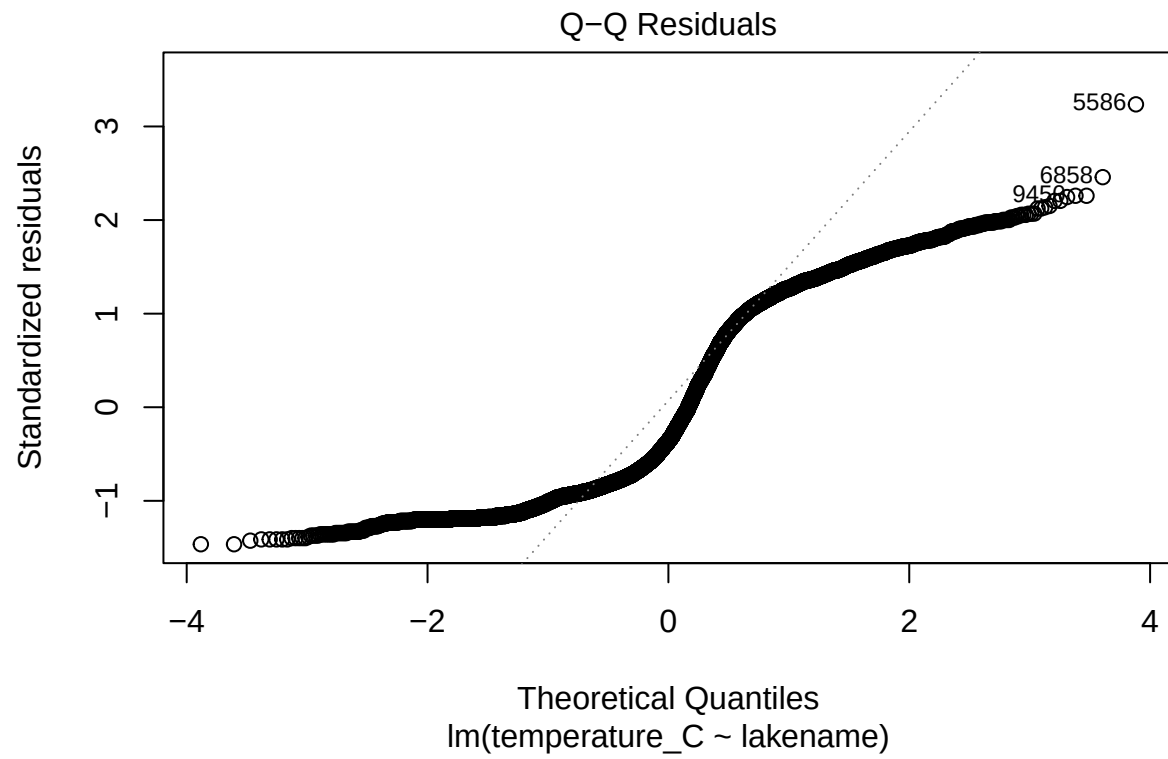
```
##
## Coefficients:
```

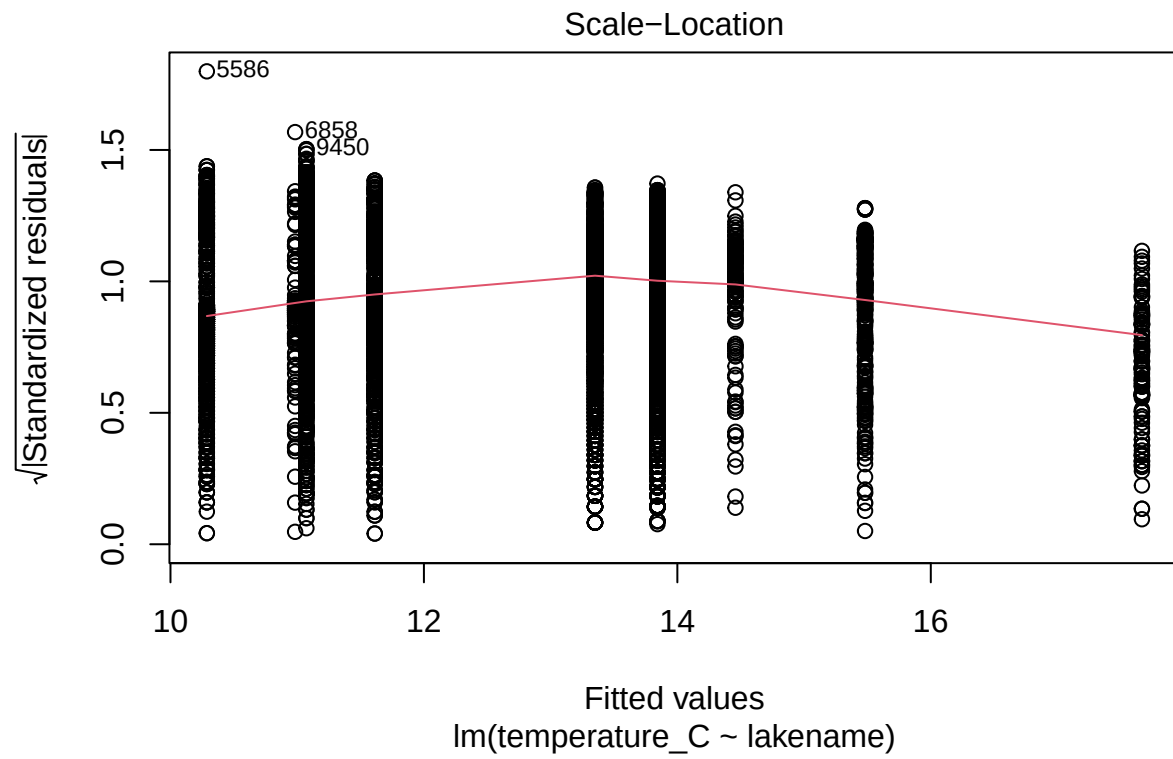
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	17.6664	0.6507	27.151	< 2e-16 ***
lakenameCrampton Lake	-2.1851	0.7565	-2.889	0.003879 **
lakenameEast Long Lake	-7.3795	0.6915	-10.671	< 2e-16 ***
lakenameHummingbird Lake	-6.6828	0.9571	-6.982	3.09e-12 ***
lakenamePaul Lake	-3.8234	0.6666	-5.735	1.00e-08 ***
lakenamePeter Lake	-4.3162	0.6652	-6.489	9.08e-11 ***
lakenameTuesday Lake	-6.5937	0.6777	-9.730	< 2e-16 ***
lakenameWard Lake	-3.2078	0.9437	-3.399	0.000679 ***
lakenameWest Long Lake	-6.0542	0.6893	-8.783	< 2e-16 ***

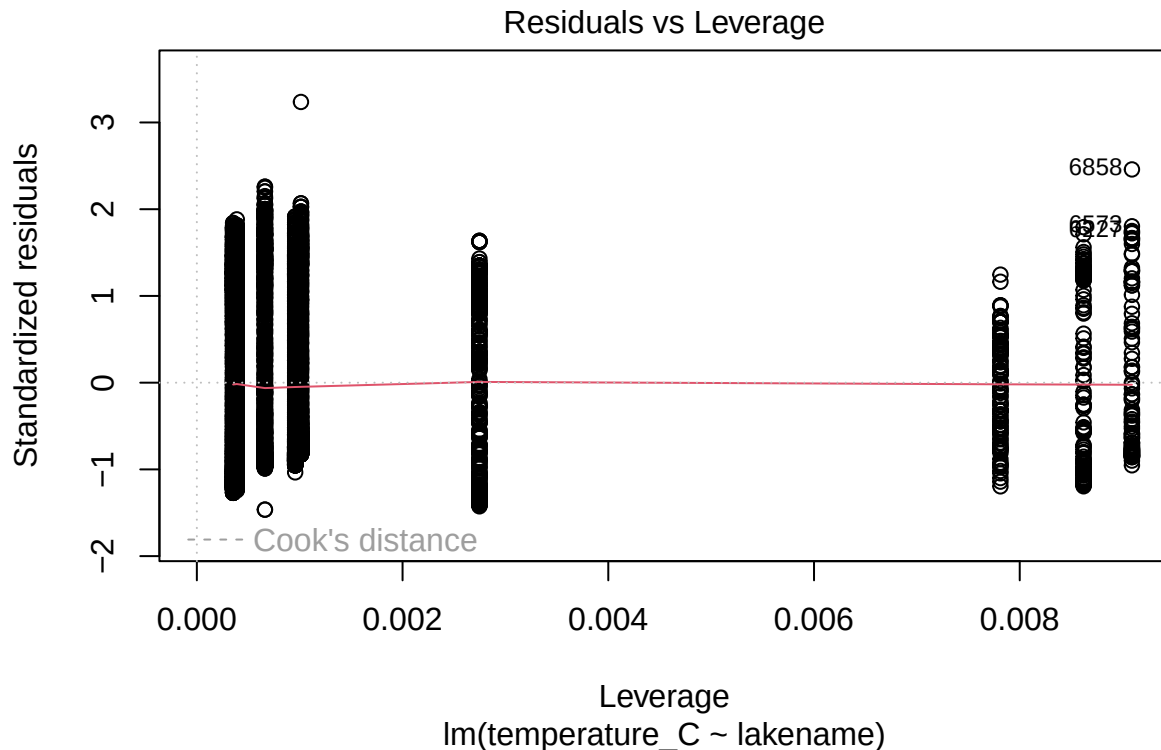
```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.361 on 9664 degrees of freedom
## Multiple R-squared:  0.04064,    Adjusted R-squared:  0.03985
## F-statistic: 51.18 on 8 and 9664 DF,  p-value: < 2.2e-16
```

```
plot(Lake_Chemistry_ANOVA2)
```









13. Is there a significant difference in mean temperature among the lakes? Report your findings.

Answer: No there is not, this is obvious by the R-squared value in the lm version of the ANOVA which shows that different lakes only account for 4.1% of variability.

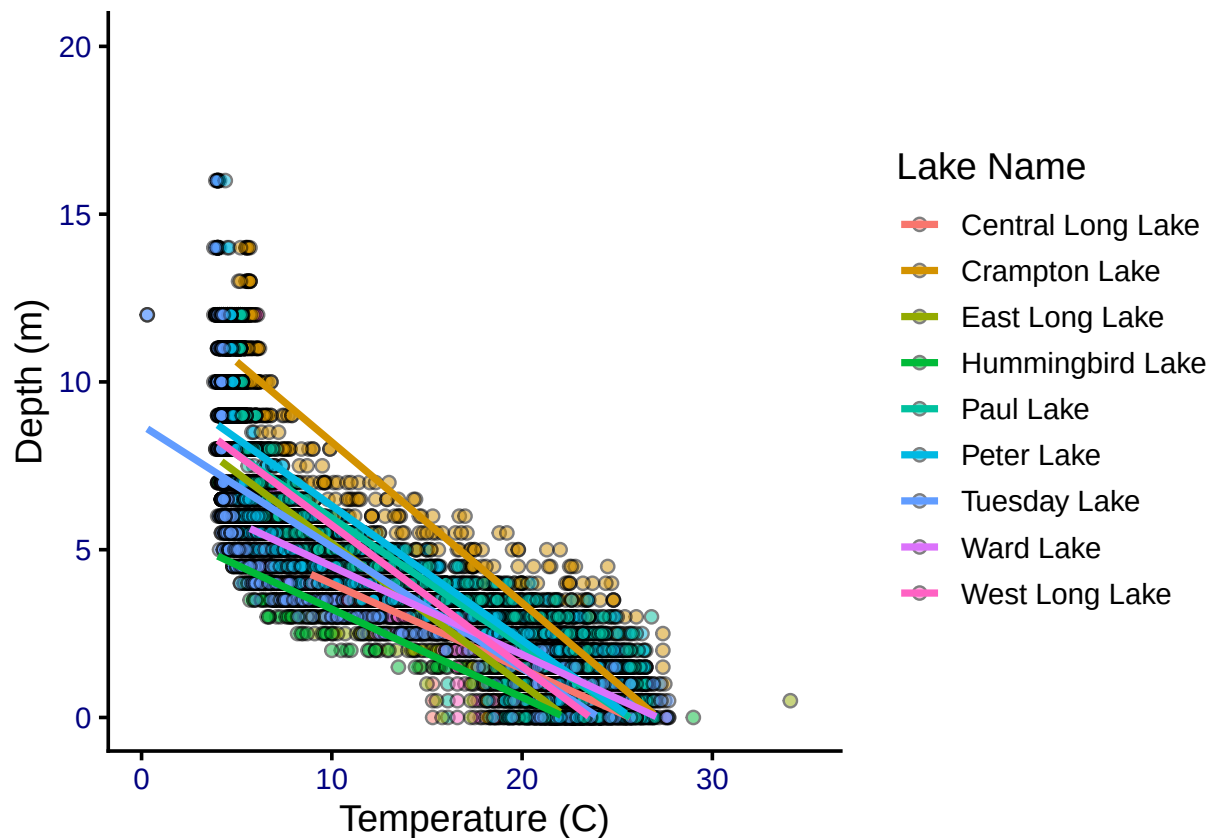
14. Create a graph that depicts temperature by depth, with a separate color for each lake. Add a `geom_smooth` (method = "lm", se = FALSE) for each lake. Make your points 50 % transparent. Adjust your y axis limits to go from 0 to 35 degrees. Clean up your graph to make it pretty.

```
#14.
temp_depth_plot2 <- ggplot(Lake_Chemistry_Sub, aes(x = temperature_C,
                                                    y = depth,
                                                    fill = lakename)) +
  geom_point(shape = 21, alpha = 0.5) +
  geom_smooth(method = "lm", se = FALSE, aes(color = lakename)) +
  labs(x = "Temperature (C)", y = "Depth (m)", color = "Lake Name",
       fill = "Lake Name") +
  xlim(0,35) +
  ylim(0,20) +
  theme(legend.position = "right")

print(temp_depth_plot2)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 94 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```



15. Use the Tukey's HSD test to determine which lakes have different means.

#15

```
TukeyHSD(Lake_Chemistry_ANOVA)
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = temperature_C ~ lakename, data = Lake_Chemistry_Sub)
##
## $lakename
##
```

	diff	lwr	upr	p adj
Crampton Lake-Central Long Lake	-2.18508757	-4.5319912	0.1618160	0.0915179
East Long Lake-Central Long Lake	-7.37946293	-9.5249061	-5.2340198	0.0000000
Hummingbird Lake-Central Long Lake	-6.68276989	-9.6520798	-3.7134600	0.0000000
Paul Lake-Central Long Lake	-3.82336936	-5.8915944	-1.7551443	0.0000004
Peter Lake-Central Long Lake	-4.31624752	-6.3799766	-2.2525184	0.0000000
Tuesday Lake-Central Long Lake	-6.59373914	-8.6961647	-4.4913136	0.0000000
Ward Lake-Central Long Lake	-3.20778556	-6.1355040	-0.2800671	0.0195251
West Long Lake-Central Long Lake	-6.05416916	-8.1927792	-3.9155591	0.0000000
East Long Lake-Crampton Lake	-5.19437536	-6.5946962	-3.7940545	0.0000000
Hummingbird Lake-Crampton Lake	-4.49768232	-6.9825914	-2.0127732	0.0000007

```
## Paul Lake-Crampton Lake      -1.63828179 -2.9171590 -0.3594045 0.0023129
## Peter Lake-Crampton Lake     -2.13115995 -3.4027534 -0.8595665 0.0000072
## Tuesday Lake-Crampton Lake   -4.40865157 -5.7421303 -3.0751729 0.0000000
## Ward Lake-Crampton Lake      -1.02269799 -3.4577560  1.4123600 0.9307880
## West Long Lake-Crampton Lake -3.86908159 -5.2589107 -2.4792525 0.0000000
## Hummingbird Lake-East Long Lake 0.69669304 -1.5988991  2.9922852 0.9905616
## Paul Lake-East Long Lake     3.55609357  2.7014024  4.4107847 0.0000000
## Peter Lake-East Long Lake    3.06321541  2.2194620  3.9069688 0.0000000
## Tuesday Lake-East Long Lake  0.78572379 -0.1486934  1.7201409 0.1828556
## Ward Lake-East Long Lake     4.17167737  1.9301428  6.4132120 0.0000003
## West Long Lake-East Long Lake 1.32529377  0.3120836  2.3385039 0.0016418
## Paul Lake-Hummingbird Lake   2.85940053  0.6358062  5.0829949 0.0021745
## Peter Lake-Hummingbird Lake  2.36652237  0.1471092  4.5859355 0.0263810
## Tuesday Lake-Hummingbird Lake 0.08903074 -2.1664094  2.3444709 1.0000000
## Ward Lake-Hummingbird Lake   3.47498433  0.4355186  6.5144501 0.0117238
## West Long Lake-Hummingbird Lake 0.62860073 -1.6606065  2.9178079 0.9952002
## Peter Lake-Paul Lake         -0.49287816 -1.1146082  0.1288519 0.2521516
## Tuesday Lake-Paul Lake       -2.77036979 -3.5104805 -2.0302590 0.0000000
## Ward Lake-Paul Lake          0.61558380 -1.5521583  2.7833259 0.9939613
## West Long Lake-Paul Lake     -2.23079980 -3.0681906 -1.3934090 0.0000000
## Tuesday Lake-Peter Lake      -2.27749162 -3.0049438 -1.5500394 0.0000000
## Ward Lake-Peter Lake         1.10846196 -1.0549910  3.2719149 0.8108720
## West Long Lake-Peter Lake    -1.73792164 -2.5641457 -0.9116976 0.0000000
## Ward Lake-Tuesday Lake       3.38595358  1.1855572  5.5863500 0.0000641
## West Long Lake-Tuesday Lake  0.53956999 -0.3790495  1.4581895 0.6673292
## West Long Lake-Ward Lake     -2.84638360 -5.0813788 -0.6113884 0.0025399
```

```
Lake_Chemistry_Groups <- HSD.test(Lake_Chemistry_ANOVA, "lakename", group = TRUE)
Lake_Chemistry_Groups
```

```
## $statistics
##      MSerror Df      Mean      CV
##    54.19142 9664 12.74849 57.74396
##
## $parameters
##      test  name.t ntr StudentizedRange alpha
##    Tukey lakename  9          4.38751 0.05
##
## $means
##               temperature_C      std      r      se Min  Max   Q25   Q50
## Central Long Lake    17.66641 4.196292  128 0.6506692 8.9 26.8 14.400 18.40
## Crampton Lake       15.48132 7.347999  364 0.3858465 5.0 27.5  7.500 17.05
## East Long Lake      10.28694 6.765204  988 0.2341999 4.2 34.1  5.000  6.50
## Hummingbird Lake    10.98364 6.779212  110 0.7018898 4.0 29.0  5.225  7.80
## Paul Lake           13.84304 7.314316 2575 0.1450697 4.7 27.7  6.500 12.40
## Peter Lake           13.35016 7.670045 2835 0.1382575 4.0 26.9  5.600 11.50
## Tuesday Lake        11.07267 7.715792 1511 0.1893795 0.3 27.7  4.400  6.80
## Ward Lake           14.45862 7.409079  116 0.6834964 5.7 27.6  7.200 12.55
## West Long Lake      11.61224 6.989346 1046 0.2276142 4.0 25.7  5.400  8.10
##
##               Q75
## Central Long Lake 21.000
## Crampton Lake    22.400
## East Long Lake    16.025
## Hummingbird Lake 16.600
```

```
## Paul Lake      21.400
## Peter Lake     21.500
## Tuesday Lake   19.400
## Ward Lake      23.200
## West Long Lake 18.800
##
## $comparison
## NULL
##
## $groups
##           temperature_C groups
## Central Long Lake    17.66641      a
## Crampton Lake        15.48132     ab
## Ward Lake            14.45862     bc
## Paul Lake            13.84304      c
## Peter Lake           13.35016      c
## West Long Lake       11.61224      d
## Tuesday Lake         11.07267     de
## Hummingbird Lake     10.98364     de
## East Long Lake       10.28694      e
##
## attr(,"class")
## [1] "group"
```

16. From the findings above, which lakes have the same mean temperature, statistically speaking, as Peter Lake? Does any lake have a mean temperature that is statistically distinct from all the other lakes?

Answer: Based on the groups, Paul and Ward Lakes have the same mean temperature. There is no lake that does not share a group with at least one other lake, so there are no statistically distinct lakes.

17. If we were just looking at Peter Lake and Paul Lake. What's another test we might explore to see whether they have distinct mean temperatures?

Answer: We could also use a two-sample T test to compare the temperature means for the two lakes.

18. Wrangle the July data to include only records for Crampton Lake and Ward Lake. Run the two-sample T-test on these data to determine whether their July temperature are same or different. What does the test say? Are the mean temperatures for the lakes equal? Does that match your answer for part 16?

```
Lake_Chemistry_CW <- Lake_Chemistry_Sub %>%
  filter(lakename %in% c("Crampton Lake", "Ward Lake"))

twosample <- t.test(Lake_Chemistry_CW$temperature_C ~ Lake_Chemistry_CW$lakename)
twosample

##
## Welch Two Sample t-test
##
## data: Lake_Chemistry_CW$temperature_C by Lake_Chemistry_CW$lakename
```

```
## t = 1.2972, df = 192.4, p-value = 0.1961
## alternative hypothesis: true difference in means between group Crampton Lake and group Ward Lake is not equal to 0
## 95 percent confidence interval:
##  -0.5323014  2.5776973
## sample estimates:
## mean in group Crampton Lake      mean in group Ward Lake
##               15.48132              14.45862
```

Answer: The test shows that the means are the same, so we accept the null hypothesis. This makes sense as the two lakes shared a group in part 16, so we would expect them to have statistically similar temperatures. We know this because the p-value is >0.05 .